

Module 6.5

Cybersecurity and Drone Security



Executive Abstract

The contemporary landscape of Unmanned Aerial Vehicles (UAVs) has transitioned from rudimentary remote-controlled platforms to sophisticated, autonomous nodes within the Internet of Unmanned Aerial Vehicles (IoUAV). This transition dictates a commensurate shift from perimeter defense to Zero Trust Architecture.

System Telemetry

Protocol:

MAVLink (Unencrypted Default)

Positioning:

GNSS (Susceptible at -128 dBm)

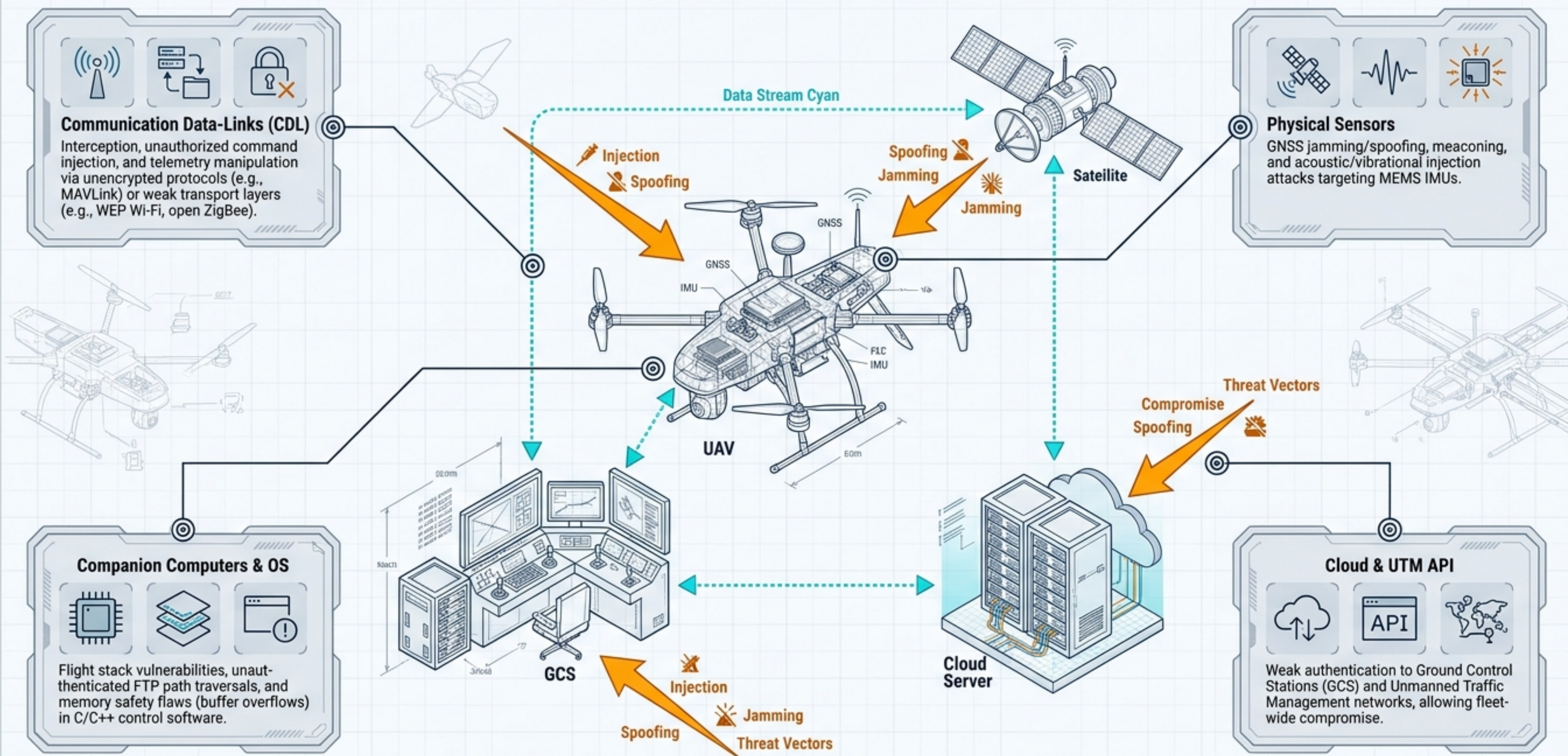
Hardware Trust:

TEE / HSM / TPM / SE required.

Threat Surface:

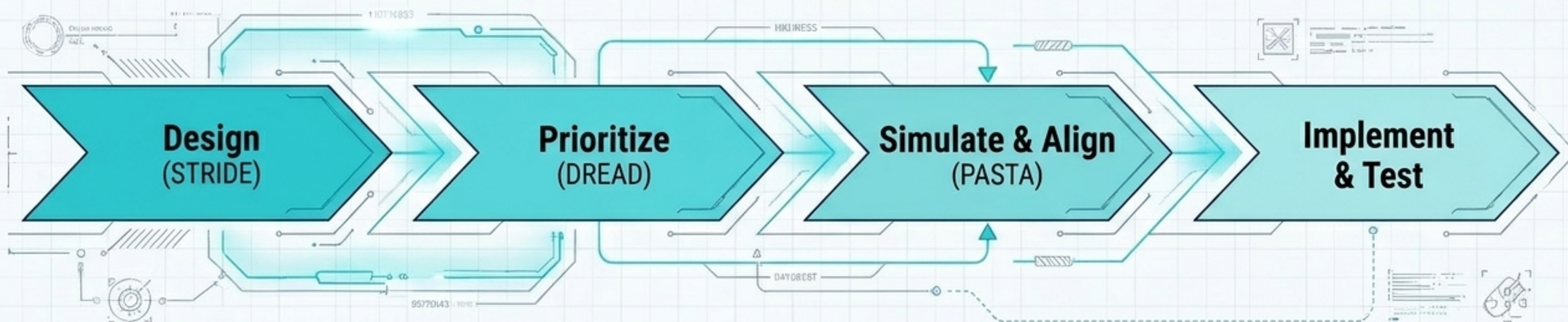
Physical, Network, Transport, and Application Layers.

The Systemic Threat Map of the IoUAV Ecosystem

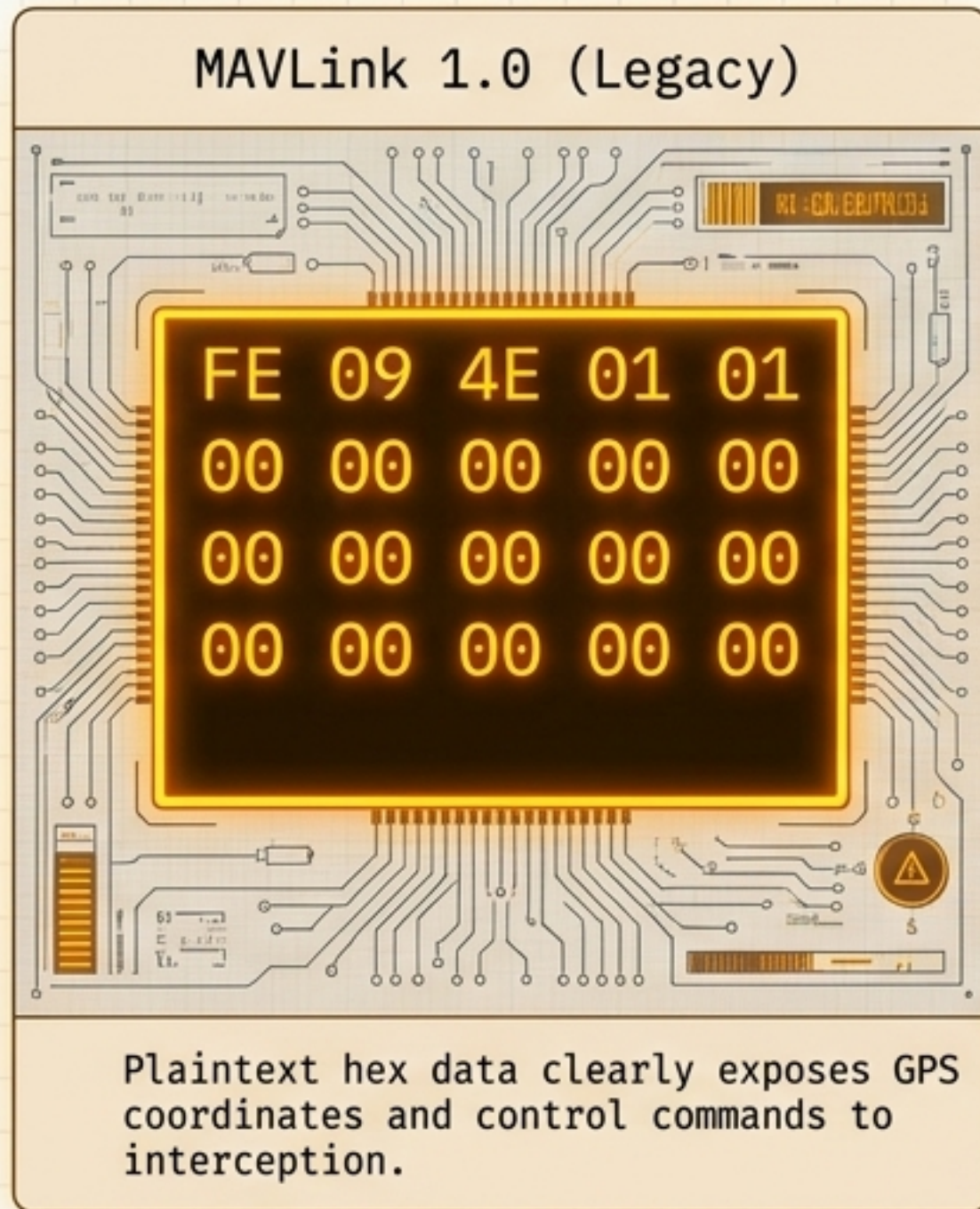


Integrating Threat Modeling Frameworks into UAV Design

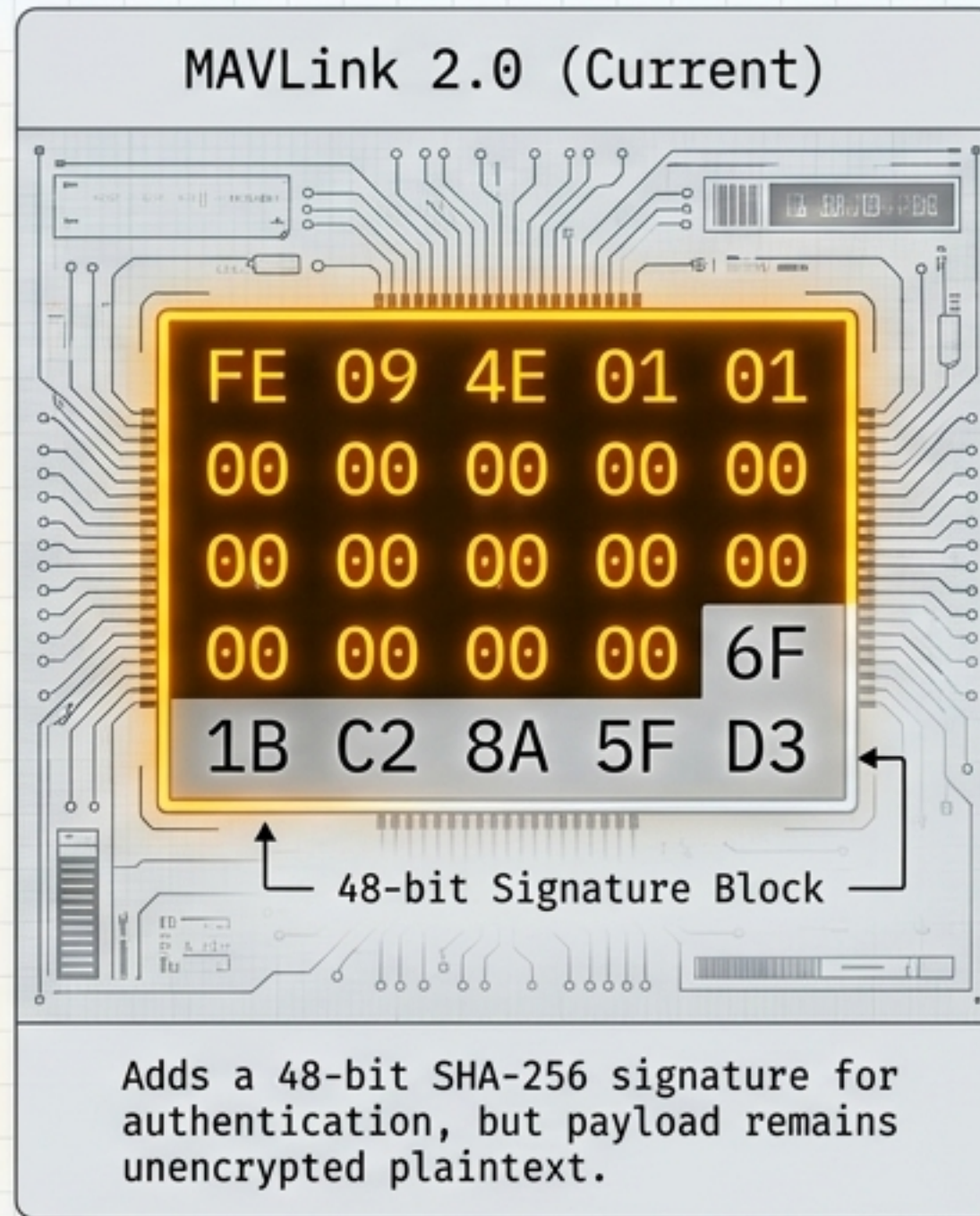
STRIDE	DREAD	PASTA
Identification	Quantification	Business Alignment
Applied during the design phase to identify threats tied to system properties. Analyzes Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege.	Used alongside STRIDE to score and rank threats. Evaluates Damage potential, Reproducibility, Exploitability, Affected Users, and Discoverability (e.g., scoring an unauthenticated RCE higher than a physical-access exploit).	A 7-step Process for Attack Simulation and Threat Analysis. Used for enterprise-level fleet operations to map technical vulnerabilities (e.g., loss of C2 link) to operational impact (e.g., payload theft, airspace violation).



Protocol Vulnerabilities and the MAVLink Plaintext Problem



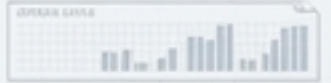
⚠️ Eavesdropping: Telemetry exposes mission parameters and edge AI models to SDR interception.



⚠️ Command Injection: Without mandated message signing, attackers forge packets to hijack flight paths.

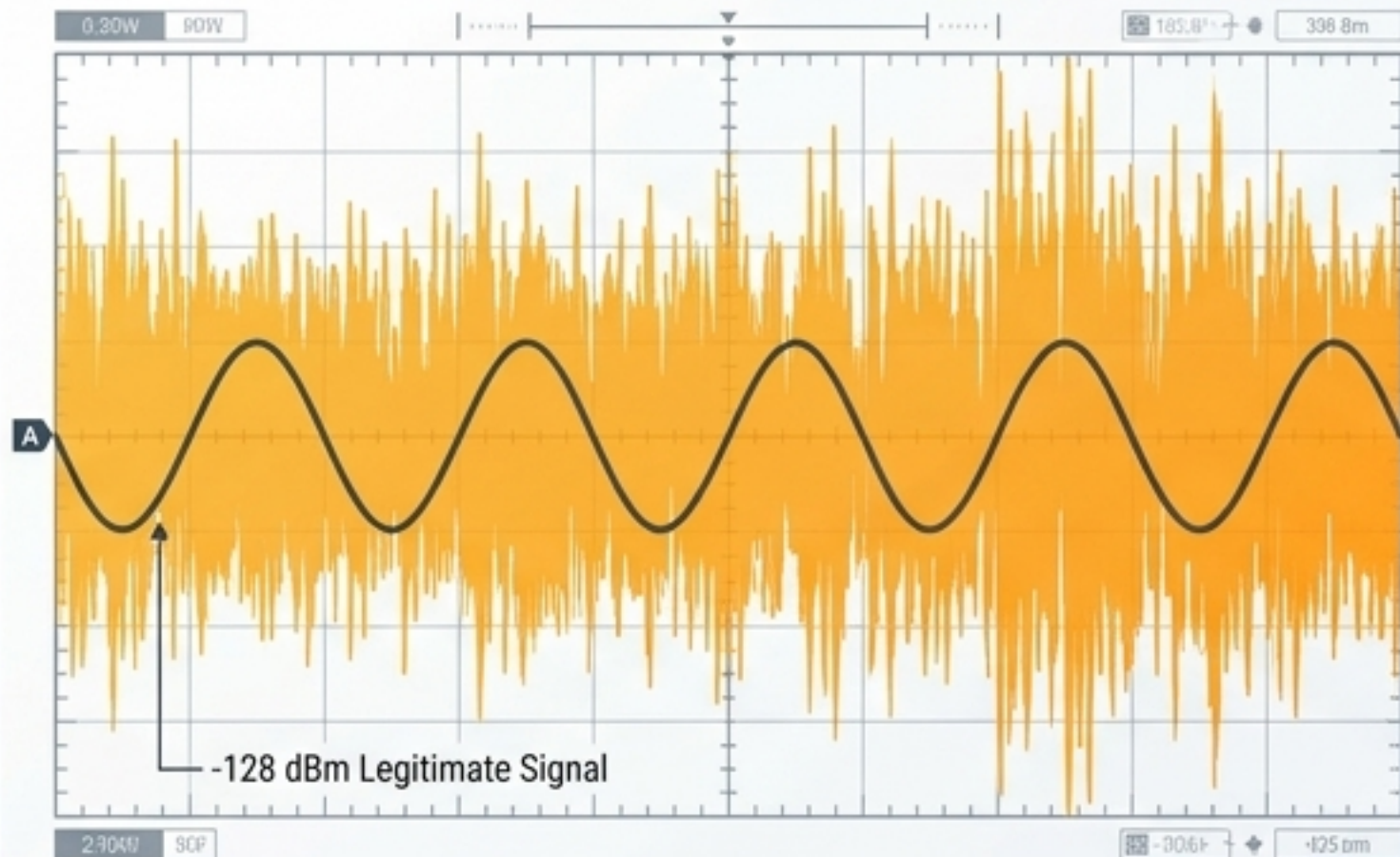


🛡️ MitM & Heartbeat Exploits: Attackers intercept unencrypted channels to manipulate critical 1-byte heartbeat messages.



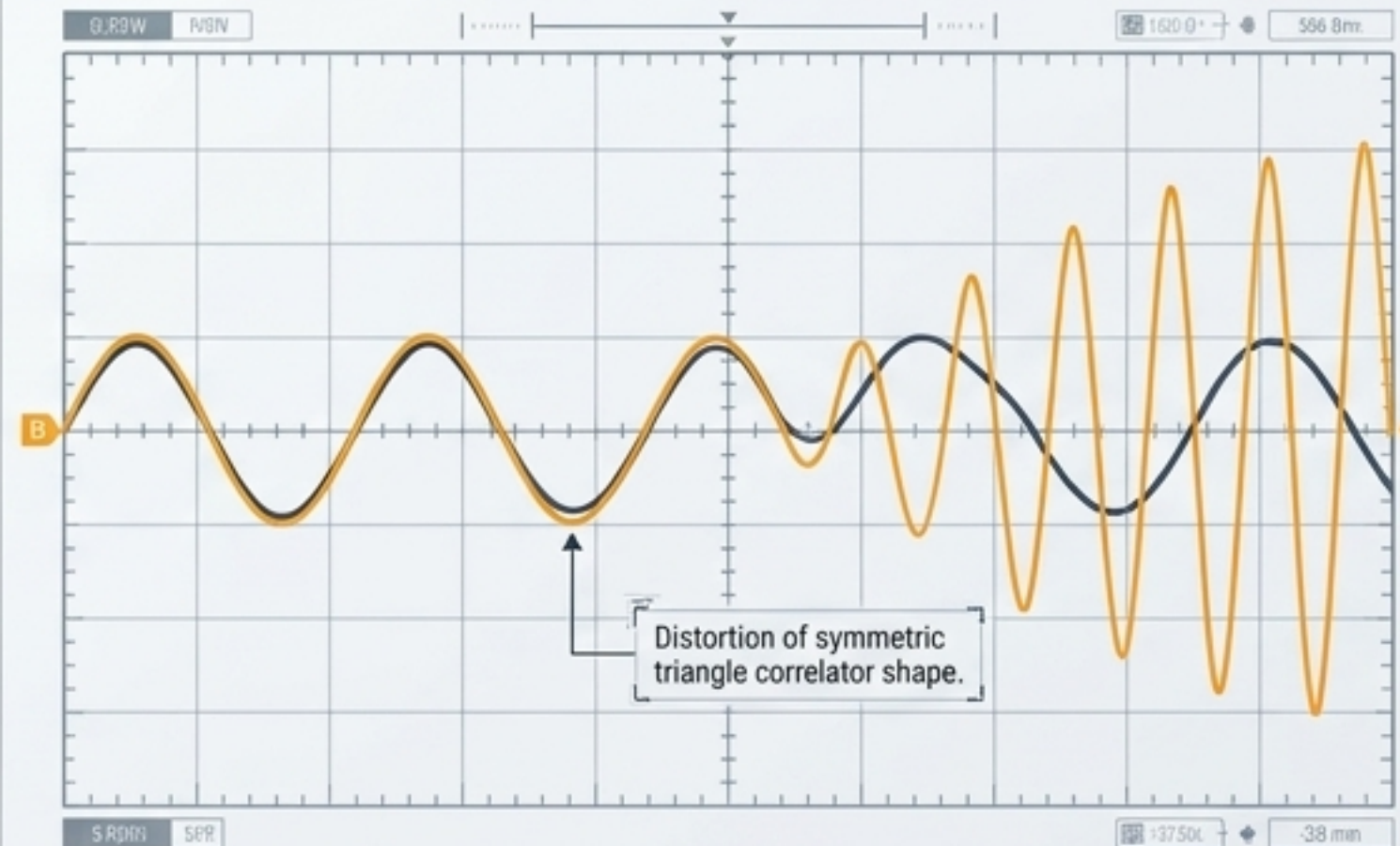
Electronic Warfare: The Physics of GNSS Interference

A Brute-Force Jamming



Overpowers civilian bands (L1/L2/L5), forcing emergency fail-safes (hover/RTH). Effectiveness measured by the J/S ratio.

B Generative Spoofing



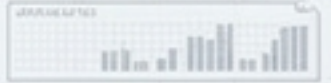
Transmitting synchronized, counterfeit signals to secretly hijack navigation. Detectable via Doppler Shifts and Correlator Shape distortion (code delay within ± 1 chip).



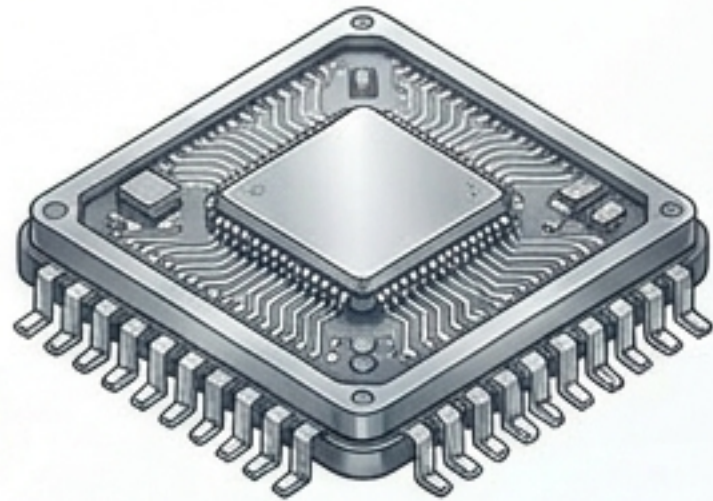
Forwarding (Meaconing)

Capturing and retransmitting genuine satellite signals with an amplified time delay. Detectable via Time Stamp Anomalies.





Anchoring Security in Hardware Roots of Trust

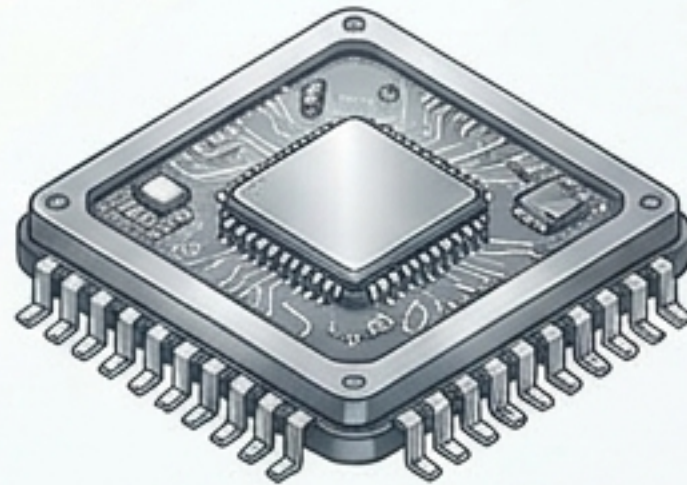


Hardware Security Module (HSM)

Physical:
Standalone appliance or PCIe card.

Security:
Provides FIPS 140-2 Level 3/4 tamper-resistance.

Use Case:
Enterprise data centers and large-scale fleet PKI key lifecycle management.

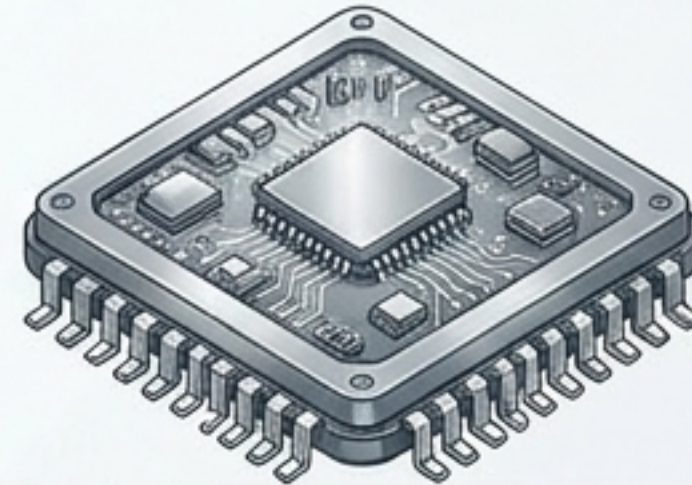


Secure Element (SE)

Physical:
Small, discrete, tamper-resistant chip.

Security:
Isolated CPU/memory independent of main system.

Use Case:
IoT-grade drones, storing biometric or payment credentials.

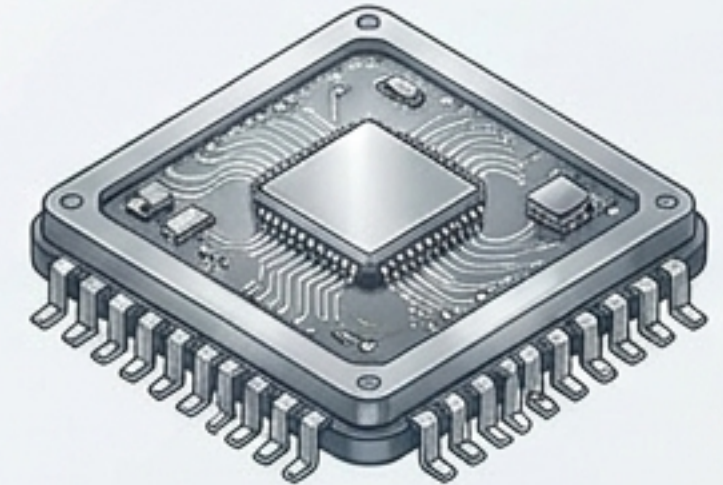


Trusted Platform Module (TPM)

Physical:
Discrete chip or integrated firmware.

Security:
Ensures platform integrity via cryptographic hashing.

Use Case:
Secure boot attestation on mission computers.



Trusted Execution Environment (TEE)

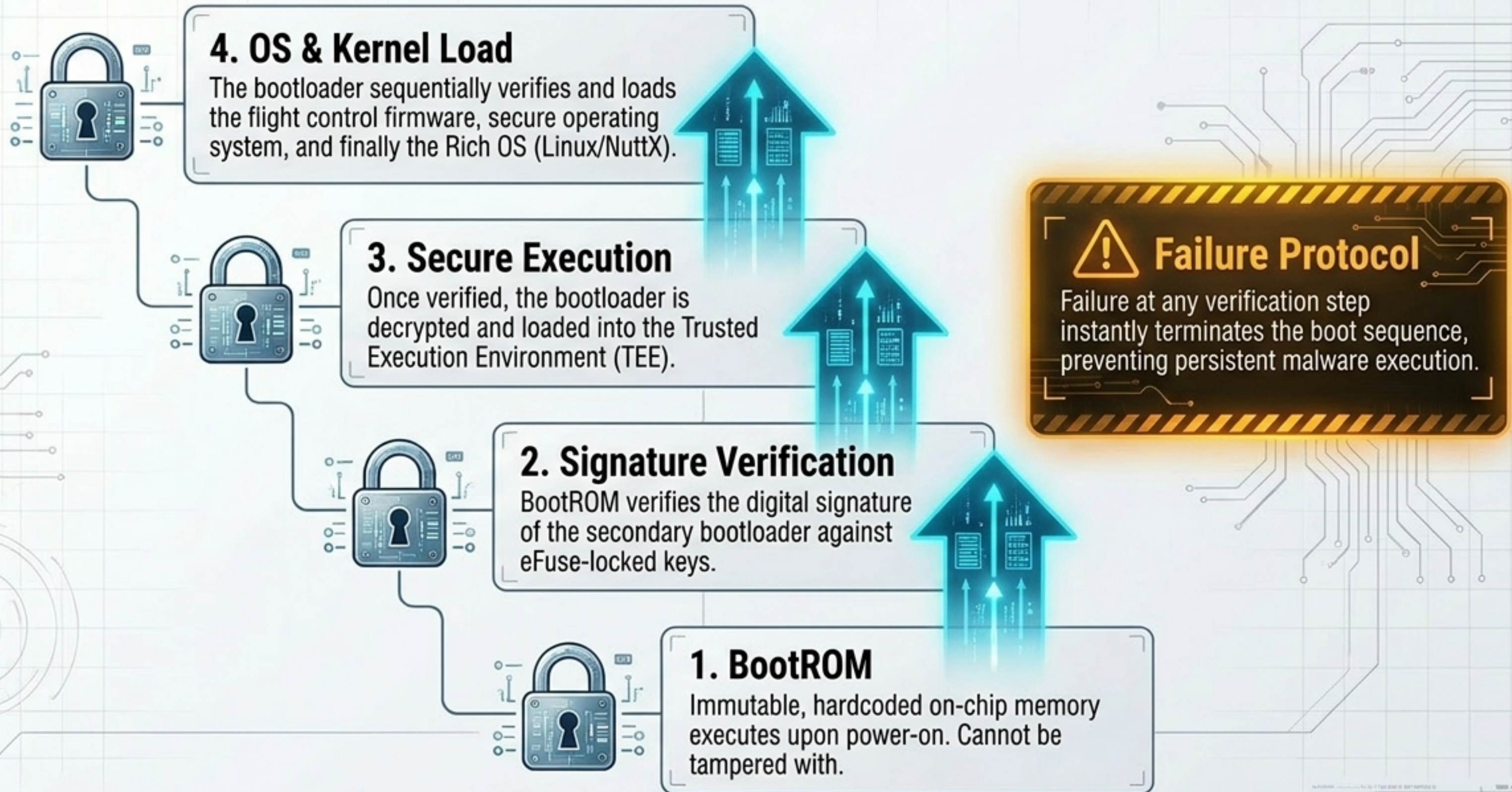
Physical:
CPU-based logical partition (e.g., ARM TrustZone).

Security:
Separates "Secure World" (crypto) from "Normal World" (Rich OS).

Use Case:
Flight controller isolation.



Cryptographic Integrity and The Secure Boot Chain





Enterprise Application Hardening and Data Privacy



OS-Level Defense (SELinux)

Enforces mandatory access control policies for all processes, operations, and files, stored in the root filesystem and protected by DM-Verity.

Runtime Protection

Employs dynamic code obfuscation, adding redundant logic to thwart reverse-engineering. White-box cryptography protects keys.

Data Isolation (Local Data Mode)

An essential privacy control that entirely disconnects the flight application from the internet, preventing sensitive telemetry transmission.

At-Rest & In-Transit Security

User data is secured via AES-256 encryption in storage, while TLS secures all cloud transmissions.

Memory Safety Exploits within the Flight Stack

CVE-2026-32706 (crsf_rc parser)



CVSS 8.1: Parser accepts an oversized, variable-length packet without bounds checking. An adjacent attacker can crash the autopilot causing total loss of flight control.

CVE-2026-32743 (MavlinkLogHandler)



CVSS 6.5: Stack-based overflow triggered via MAVLink FTP log requests. The `scanf` function parses paths without a width specifier, crashing the MAVLink task and causing DoS.

Failure Protocol

CRITICAL FAILURE: Instantaneous termination of compromised process, preventing escalating impact.

Applied Cryptography: Overcoming MAVLink Overhead



The Encryption Dilemma

Standard MAVLink eschews encryption to preserve low latency. AES-CTR, ChaCha20, Speck-CTR, and Rabbit ciphers were evaluated on live PX4/ArduPilot testbeds.

The MAVShield Cipher Architecture



Proven robust against differential cryptanalysis, passing 24 distinct NIST/Diehard attacks while providing comprehensive confidentiality with negligible overhead.

Cyber-Physical Exploits: Acoustic and Inertial Attacks

Resonant Frequency Injection

⚠️ Attackers target the specific physical resonance frequencies of MEMS sensors (typically between 15-25 kHz) using directed acoustic or vibrational equipment.

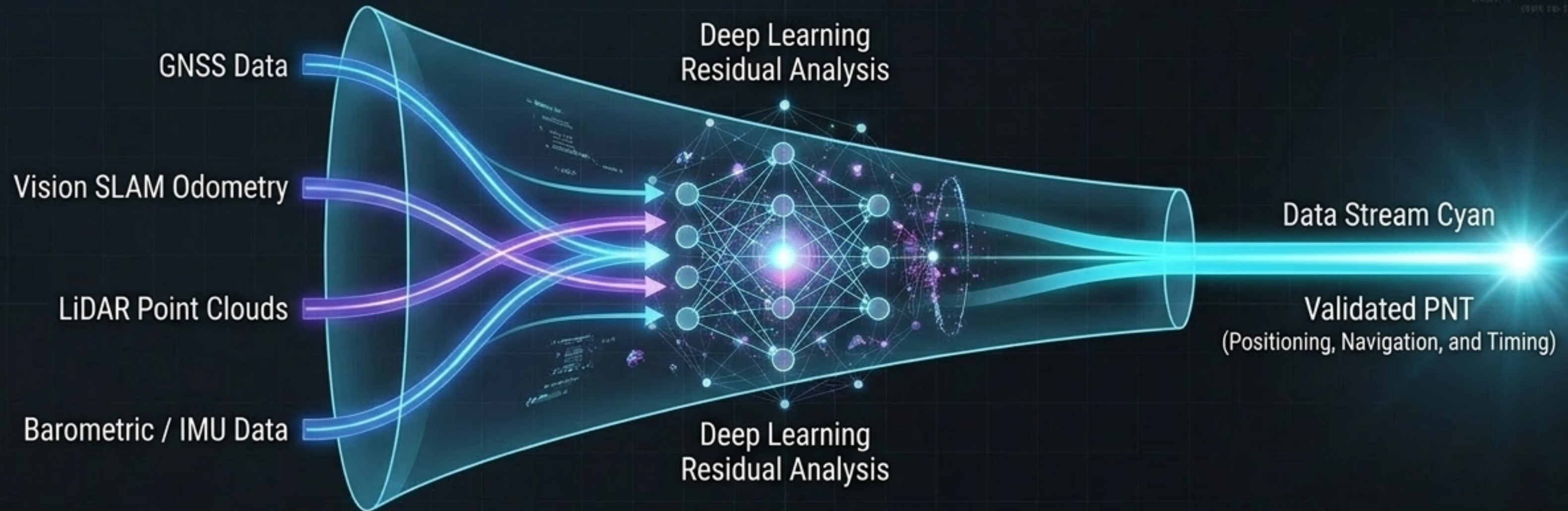
Phantom Acceleration

⚠️ Targeted soundwaves induce false mechanical signals that the drone's flight controller interprets as actual movement, injecting severe angular-rate biases.

Coupled System Failure

⚠️ In loosely coupled GNSS/INS architectures, these injected biases accumulate over time. If synchronized with GNSS spoofing, traditional fault detection mechanisms collapse, allowing seamless trajectory hijacking.

Defeating Spoofing via AI-Assisted Sensor Fusion



Cross-Verification

Relieving exclusive reliance on GNSS by cross-referencing inertial data with visual odometry and 5G positioning.

Innovation Sequence

Monitoring the difference between predicted and measured values. Statistical hypothesis testing triggers alerts when residuals exceed expected bounds.

Signal Watermarking

Analyzing the entropy of incoming signals to verify their cryptographic authenticity before integration, defeating sophisticated spoofing.

Zero Trust Architecture in UAS Traffic Management (UTM)

The Post-Perimeter Paradigm

⚠ Traditional perimeter defense is obsolete in highly networked loUAV ecosystems. ZTA assumes the network is perpetually hostile.

Micro-segmentation

⚠ Restricting lateral movement. A compromised companion computer cannot access the flight controller's PID loop without re-authentication.



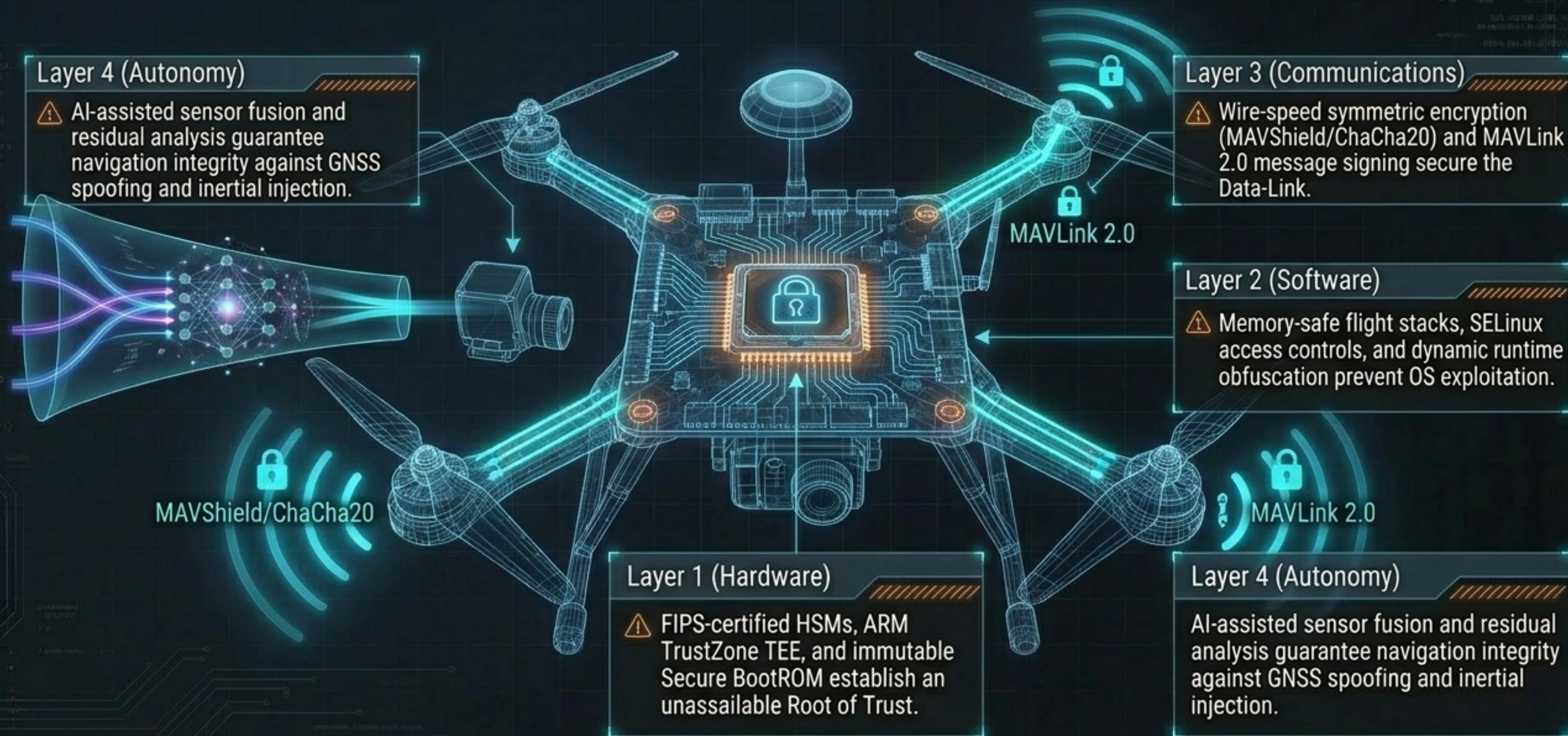
Continuous Authentication

Devices must continuously prove their identity and integrity (via Measured Boot attestation tokens) to the UTM, rather than authenticating only upon connection.

Dynamic Trust Monitoring

⚠ Ground stations evaluate drone behavior (trajectory, energy use) in real-time to calculate dynamic trust scores, instantly isolating anomalous nodes.

Synthesis: The Complete Defense-in-Depth Paradigm



The Future of IoUAV Engineering



As Unmanned Aerial Systems evolve into critical national infrastructure, cybersecurity can no longer be an overlay—it must be a foundational engineering prerequisite.

The Mandate

Future deployment requires mandatory hardware acceleration for cryptographic workloads, standardized post-quantum resilience, and physical tamper-proofing across all commercial and defense platforms.

Reference

Data and benchmarks derived from recent architectural analyses of PX4 Autopilot CVEs, MAVSec implementations, and DJI Enterprise White Paper specifications.